

3. A battery has an emf of 12 V.

- (a) (i) Explain what this statement means. [2]

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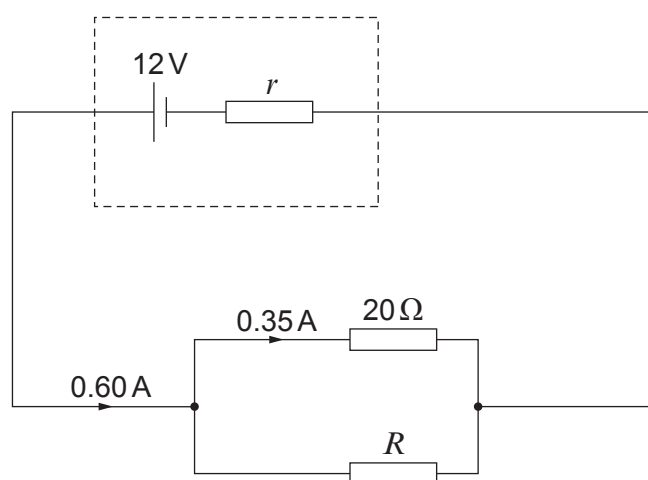
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- (ii) State why it is usually a disadvantage for a battery to have an internal resistance. [1]

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- (b) The battery is included in the circuit shown.



- (i) Use the information shown on the circuit diagram to calculate:

- I. the resistance,  $R$ . [2]

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- II. the internal resistance,  $r$ . [2]

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- (ii) I. Calculate the energy transferred in the  $20\ \Omega$  resistor in 40 minutes. [2]

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- II. State the energy transfer that occurs in the  $20\ \Omega$  resistor and explain, in terms of the motion of free electrons, how it takes place. [2]

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- (iii) Dafydd claims that if  $R$  is replaced by a higher resistance, but no other changes are made, the power dissipation **in the  $20\ \Omega$  resistor** will increase. Evaluate this claim, showing clear reasoning. [3]

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